A Project Report on

SMART INTERVIEW ASSISTANT: AN AI EVALUATION AND FEEDBACK SYSTEM

submitted in partial fulfillment of the requirements for the award of the degree of

BACHELOR OF ENGINEERING

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Abstract

In today's competitive job market, interviews play a pivotal role in assessing candidates' technical knowledge, communication skills, and overall readiness. However, traditional interview processes are often subjective, time-consuming, and prone to human bias, leading to inconsistent evaluation outcomes. To address these challenges, the Smart Interview Assistant introduces an AI-powered evaluation and feedback system that automates the assessment process using Natural Language Processing (NLP), speech analysis, and transformer-based models, thereby contributing to SDG 9 – Industry, Innovation, and Infrastructure through the adoption of advanced AI technologies.

The system analyzes candidates' responses in real time by converting speech to text and applying semantic and contextual analysis to evaluate parameters such as confidence, clarity, and response relevance. It leverages advanced machine learning and large language models to score candidates based on communication quality, content accuracy, and behavioral indicators derived from speech patterns. This enables a more structured and objective assessment compared to traditional methods, aligning with SDG 8 – Decent Work and Economic Growth by enhancing fair hiring practices and workforce readiness.

Additionally, the integrated feedback module generates personalized and constructive feedback by highlighting strengths and identifying areas for improvement, supporting continuous learning and skill development in line with SDG 4 – Quality Education. The system also incorporates speech-based features such as tone, pauses, and speaking rate to enhance the evaluation of communication effectiveness. By providing a standardized, data-driven evaluation framework, the Smart Interview Assistant improves recruitment efficiency, reduces bias, and promotes equal opportunities, thereby contributing to SDG 10 – Reduced Inequalities.

Furthermore, the development and deployment of such intelligent systems encourage collaboration between academia, industry, and technology providers, reflecting the principles of SDG 17 – Partnerships for the Goals. Ultimately, this system transforms traditional interviewing into a more scalable, transparent, and intelligent process aligned with modern AI-driven technologies and global sustainable development objectives.

Keywords: Speech analysis, Interview automation, Real-time Feedback system, Deep learning, Communication assessment, Performance analysis.

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ABBREVAION

AI	Artificial Intelligence
ML	Machine Learning
NLP	Natural Language Processing
LLM	Large Language Model
ASR	Automatic Speech Recognition
API	Application Programming Interface
UI	User Interface
DFD	Data Flow Diagram
SDG	Sustainable Development Goal
CNN	Convolutional Neural Network
BERT	Bidirectional Encoder Representations from Transformers
GPT	Generative Pre-trained Transformer
LTS	Long Term Support
NoSQL	Not Only SQL (Database Type)
HTTP	HyperText Transfer Protocol
GPU	Graphics Processing Unit
CPU	Central Processing Unit

Chapter 1

Introduction

1. Introduction

In today's highly competitive job market, interviews have become a critical stage in assessing a candidate's communication skills, technical knowledge, and overall personality. However, traditional interview processes often face challenges such as scheduling constraints, subjective evaluations, and the lack of structured feedback mechanisms. These limitations not only reduce the efficiency of recruiters but also leave candidates without clear insights into their performance or areas for improvement.

Recent advancements in Artificial Intelligence (AI), Natural Language Processing (NLP), and transformer-based models have opened new possibilities for transforming the recruitment process. Modern AI systems can automate interview workflows, generate context-aware and role-specific questions, and provide objective, data-driven feedback based on candidate responses [1], [3], [4]. These technologies help reduce human bias, improve consistency in evaluation, and enhance the overall interview experience for both recruiters and candidates.

The proposed Smart Interview Assistant is designed to address these challenges by integrating AI-powered question generation, resume parsing, speech-to-text transcription, and real-time response analysis. The system supports multiple interview types, including HR and technical rounds, and evaluates key parameters such as communication clarity, response relevance, and confidence level using speech and NLP-based techniques [13], [14]. This enables a structured and scalable approach to interview assessment, helping recruiters save time while allowing candidates to improve their interview readiness through continuous feedback and performance insights.

1.1 Sustainable Development Goals (SDGs)

1. **SDG 4 – Quality Education:**
   - Enhances candidates' learning and skill development through AI-driven personalized feedback systems.
   - Acts as a self-improvement platform that helps individuals refine communication and technical abilities for improved employability.
2. **SDG 8 – Decent Work and Economic Growth:**
   - Promotes fair and efficient hiring by minimizing bias through automated and data-driven evaluation methods.
   - Assists organizations in identifying suitable talent, thereby improving workforce quality and productivity.
3. **SDG 9 – Industry, Innovation, and Infrastructure:**
   - Utilizes AI, NLP, and modern computing technologies to modernize traditional recruitment systems.
   - Encourages innovation by building scalable and intelligent digital infrastructure for hiring processes.
4. **SDG 10 – Reduced Inequalities:**
   - Supports unbiased evaluation irrespective of gender, background, or region through standardized assessment techniques.
   - Provides equal opportunities for candidates from diverse educational and socio-economic backgrounds.
5. **SDG 17 – Partnerships for the Goals:**
   - Encourages collaboration between academic institutions and industry through AI-driven solutions.

- Supports collective advancement toward sustainable and technology-enabled employment ecosystems.

1.2 Motivation

Traditional interview methods often lack standardization and objectivity, resulting in inconsistent evaluations and difficulty in identifying the most suitable candidates. The motivation behind this project arises from the need to improve the efficiency, transparency, and fairness of the interview process using advanced AI technologies.

By integrating Natural Language Processing, speech analysis, and machine learning techniques, the Smart Interview Assistant minimizes human bias and provides structured evaluation metrics. It enables candidates to receive meaningful, real-time feedback, helping them improve their communication skills and confidence over time. This approach not only enhances recruiter productivity but also creates a more transparent, scalable, and data-driven hiring process aligned with modern industry requirements [1], [4].

1.3 System Design

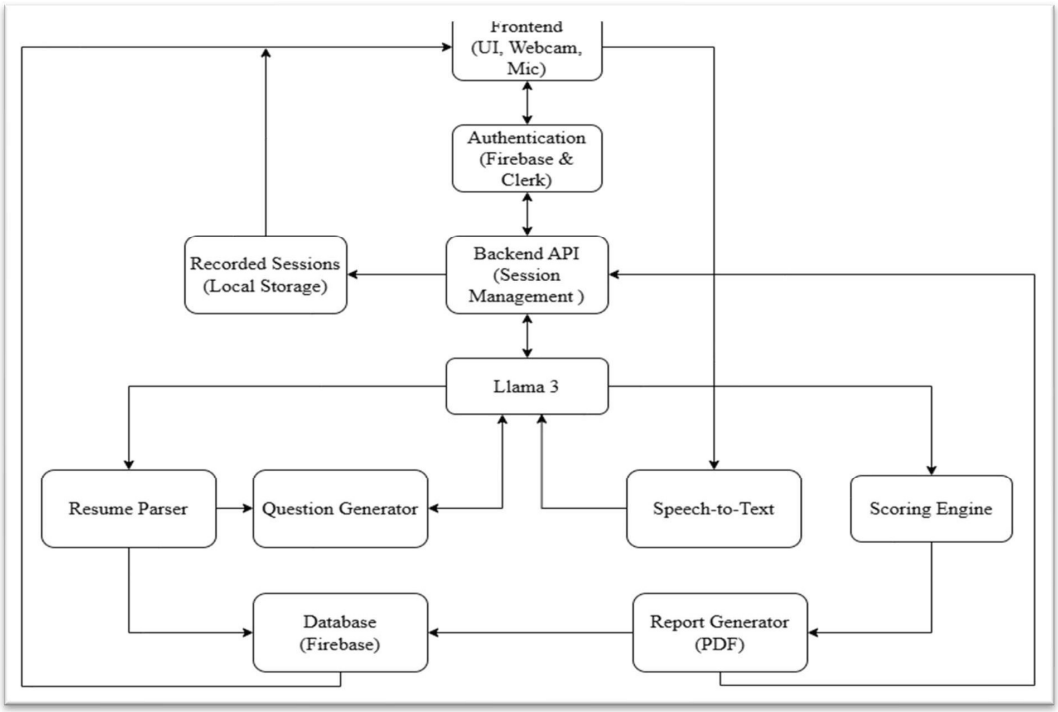

Fig 1.3. 1: System Design

The system design of the Smart Interview Assistant illustrates the overall workflow and interaction between different components, including the user interface, backend services, AI modules, and data storage systems. As shown in Fig. 1.3, the frontend acts as the primary interaction layer where users provide inputs through webcam and microphone. User authentication and access control are managed using Firebase and Clerk, ensuring secure and reliable communication between system components.

The backend layer is responsible for handling API requests, managing interview sessions, and coordinating with AI modules for processing user responses. Core components such as the Resume Parser, Question Generator, Speech-to-Text module, and Evaluation Engine work together to enable intelligent interview execution. Transformer-based models such as LLaMA 3 are utilized for generating context-aware interview questions and performing semantic analysis of candidate responses, allowing accurate evaluation based on relevance, clarity, and communication quality [1], [3], [4].

The processed data is securely stored in Firebase services, including Firestore for structured data and Firebase Storage for media files such as audio and video recordings. A dedicated Report Generator module compiles evaluation results and generates detailed performance reports for users. This modular and scalable architecture ensures efficient data flow, real-time processing, and seamless integration of AI-driven functionalities, making the system suitable for both academic and organizational use [13], [14].

Chapter 2

Literature Survey/ Existing System

2. Survey/ Existing System

2.1 Overview of Existing Research

Table 2.1.1: Literature Survey

Year	Title	Domain / Focus	Techniques Used	Key Contribution / Gap
2024	Large Language Models for Automated Evaluation: A Survey	Interview evaluation	LLM, NLP	Inconsistent evaluation outputs
2023	AI-based Interview Evaluation Systems: A Review	Interview automation	NLP, ML	Limited personalization and feedback
2023	AI in Hiring: Ethical and Bias Considerations	AI fairness	Survey, bias detection	Lack of transparency
2021	Speech Emotion Recognition: Recent Advances	Speech analysis	Deep learning	Sensitive to noise and accents

Year	Title	Domain / Focus	Techniques Used	Key Contribution / Gap
2019	Multimodal Machine Learning: A Survey and Taxonomy	Multimodal AI	Deep learning	High computation cost
2020	Sentiment Analysis Using Deep Learning: A Survey	NLP analysis	Deep learning	Limited contextual understanding
2020	Language Models are Few-Shot Learners	LLM	Transformer models	High computational cost
2019	BERT: Pre-training of Deep Bidirectional Transformers	NLP	Transformer models	Requires fine-tuning
2021	AI in Talent Assessment: A Review	HR tech	AI-based evaluation	Lack of real-time feedback
2021	Deep Learning for NLP: A Survey	NLP	Neural networks	Domain dependency issues
2022	Multimodal Emotion Recognition Using Deep Learning	Emotion analysis	CNN, multimodal	Complex integration
2022	Data2Vec: Self-Supervised Learning Framework	Speech/NLP	Self-supervised learning	High training complexity

2.2 Detailed Analysis of Key Research Papers

[6] Large Language Models for Automated Evaluation: A Survey, 2024.

This study explores the application of large language models in automated evaluation systems, particularly focusing on their ability to generate human-like responses and assess textual inputs. The paper highlights how transformer-based architectures improve contextual understanding and enable automated scoring mechanisms. However, it also identifies challenges such as inconsistent outputs and lack of controlled evaluation standards. This work is highly relevant to the Smart Interview Assistant as it forms the foundation for using LLMs in response evaluation and feedback generation.

[7] AI-based Interview Evaluation Systems: A Review, IEEE, 2023.

This paper provides a comprehensive overview of AI-driven interview evaluation systems, focusing on the use of NLP and machine learning techniques for automated candidate assessment. It discusses various approaches for analyzing candidate responses and generating scores based on predefined metrics. The study highlights limitations such as lack of personalization and absence of real-time feedback. These insights directly influence the design of the Smart Interview Assistant, which aims to provide dynamic question generation and instant feedback.

[13] Speech Emotion Recognition: Recent Advances, IEEE, 2021.

This research focuses on speech-based emotion recognition using deep learning techniques and acoustic feature extraction. It demonstrates how speech characteristics such as tone, pitch, and pauses can be used to assess emotional states and confidence levels. Despite its effectiveness, the study identifies challenges related to noise sensitivity and accent variations. This work supports the Smart Interview Assistant in incorporating speech analysis for evaluating communication quality and confidence.

[11] Multimodal machine learning: A survey and taxonomy, IEEE TPAMI, 2019.

This paper presents a comprehensive survey of multimodal machine learning techniques that integrate multiple data sources such as text, speech, and visual inputs. It explains how combining these modalities leads to more accurate and robust predictions. However, it also highlights the complexity and computational cost associated with multimodal systems. This research provides a conceptual foundation for extending the Smart Interview Assistant toward multi-dimensional evaluation.

[15] Sentiment Analysis Using Deep Learning: A Survey, 2020.

This study examines deep learning approaches for sentiment analysis in natural language processing. It discusses techniques for extracting semantic meaning and emotional context from textual data using neural networks. While effective, the paper points out limitations in contextual understanding across different domains. This work contributes to the Smart Interview Assistant by enabling semantic evaluation of candidate responses and improving feedback generation.

Chapter 3

Limitation of Existing System

3. Limitation of Existing System

Despite significant advancements in Artificial Intelligence, Natural Language Processing, and automated evaluation systems, several critical research gaps remain in the domain of AI-driven interview assessment. Existing studies have explored individual components such as NLP-based response analysis, speech emotion recognition, and multimodal evaluation techniques; however, most systems operate in isolation and fail to provide a unified and comprehensive solution. A major gap lies in the lack of integrated end-to-end systems that combine multiple modalities such as text, speech, and contextual understanding into a single evaluation framework. Although multimodal approaches have been proposed, they often suffer from high computational complexity and limited real-world deployment, leading to systems that focus only on specific aspects rather than providing holistic candidate assessment [11], [12].

Moreover, issues related to bias, fairness, scalability, and accessibility continue to hinder the effectiveness of current systems. AI-driven hiring tools may unintentionally introduce bias due to imbalanced training data or algorithmic limitations, raising concerns about transparency and fairness in evaluation [8], [9]. At the same time, many advanced systems require significant computational resources, making them difficult to deploy in academic or small-scale environments. Addressing these gaps, the Smart Interview Assistant proposes an integrated, real-time, and context-aware evaluation framework that combines NLP, speech analysis, and transformer-based models. By focusing on scalability, fairness, and structured feedback generation, the system aims to provide a reliable, efficient, and practical solution for modern interview evaluation.

- **Bias in Evaluation:**
  Many AI-driven interview platforms suffer from bias related to gender, accent, or background, which can affect fairness in candidate assessment. Studies on AI hiring systems highlight the presence of algorithmic bias and lack of transparency in decision-making processes [8], [9].
- **High Cost of Implementation:**
  Most commercial AI interview solutions require significant infrastructure, computational resources, and licensing costs, making them inaccessible for small organizations and academic institutions. The use of advanced models such as large language models further increases deployment and maintenance costs [6].
- **Limited and Generic Feedback:**
  Existing systems often provide only basic or generic feedback, lacking detailed insights into candidate performance. Research indicates that effective automated feedback must be structured, personalized, and actionable to be useful for candidates [5], [7].
- **Lack of Integrated End-to-End Solutions:**
  Many available tools focus on isolated functionalities such as speech transcription or resume screening rather than offering a complete interview evaluation system. This lack of integration reduces overall system efficiency and usability in real-world scenarios [7].
- **Insufficient Context-Aware Evaluation:**
  Traditional systems struggle with understanding contextual meaning in candidate responses, leading to inaccurate evaluation of answers. Although NLP models have improved significantly, challenges related to domain adaptation and deep contextual understanding still remain [3], [15].

Chapter 4

Problem Statement and Objective

4. Problem Statement and Objective

4.1 Problem Statement

The traditional interview process is time-consuming, prone to human bias, and often lacks detailed analytical feedback. While several AI-powered tools have attempted to address parts of this challenge, they mostly provide partial solutions and fail to deliver a comprehensive, transparent, and affordable platform. There remains a significant gap in creating a system that can seamlessly generate resume-based and role-specific interview questions, conduct AI-driven interviews with real-time transcription, provide objective feedback on both verbal and non-verbal cues, and finally deliver a structured performance report that benefits both candidates and recruiters.

4.2 Objective Of the Project

- **Design an AI-Powered Interview Platform:**
  The primary objective is to develop an intelligent interview platform that leverages Artificial Intelligence to automate and enhance the interview process, ensuring consistency and efficiency in evaluation.
- **Develop a Resume Parser for Personalized Question Generation:**
  The system aims to implement a resume parsing module that extracts key candidate information such as skills, experience, and education, enabling the generation of context-aware and role-specific interview questions.
- **Implement Real-Time Feedback Mechanisms:**
  Another objective is to provide instant and actionable feedback to candidates during interview sessions using NLP and speech analysis techniques, helping them improve their performance in real time [5], [7].
- **Ensure Fairness, Transparency, and Affordability:**
  The system focuses on minimizing bias and maintaining transparency in evaluation by using standardized AI-based scoring methods, while also ensuring cost-effective deployment [8], [9].
- **Integrate Analytics Dashboard for Recruiters and Candidates:**
  To design an interactive dashboard that visualizes performance metrics, interview insights, and progress tracking, enabling data-driven decision-making.

Chapter 5

Proposed System

5. Proposed System

The Smart Interview Assistant is designed as an AI-driven system that automates the interview process by integrating Natural Language Processing, speech analysis, and transformer-based models. The system aims to provide a structured and objective evaluation framework that overcomes the limitations of traditional interview methods. It enables automated question generation, real-time response analysis, and feedback generation, ensuring consistency and efficiency in candidate assessment.

The system follows a modular architecture consisting of frontend, backend, AI processing, and data storage components. Users interact with the system through a web-based interface, where they can upload resumes or select job roles to initiate interview sessions. Based on the provided input, the system dynamically generates personalized interview questions using transformer-based models, ensuring context-aware and role-specific interactions [1], [3], [4].

During the interview session, user responses are recorded through audio input and processed using speech-to-text conversion techniques. The system then applies NLP-based semantic analysis and speech feature extraction to evaluate response quality. Key parameters such as relevance, clarity, coherence, and confidence are assessed to provide a comprehensive evaluation of candidate performance [13], [14].

The processed data is further used to generate real-time feedback and detailed post-interview reports, helping candidates identify strengths and areas for improvement. The system ensures scalability and secure data handling through cloud-based storage and efficient backend services. Overall, the proposed system provides a reliable, intelligent, and user-friendly solution for modern interview evaluation.

5.1 System Design and Architecture

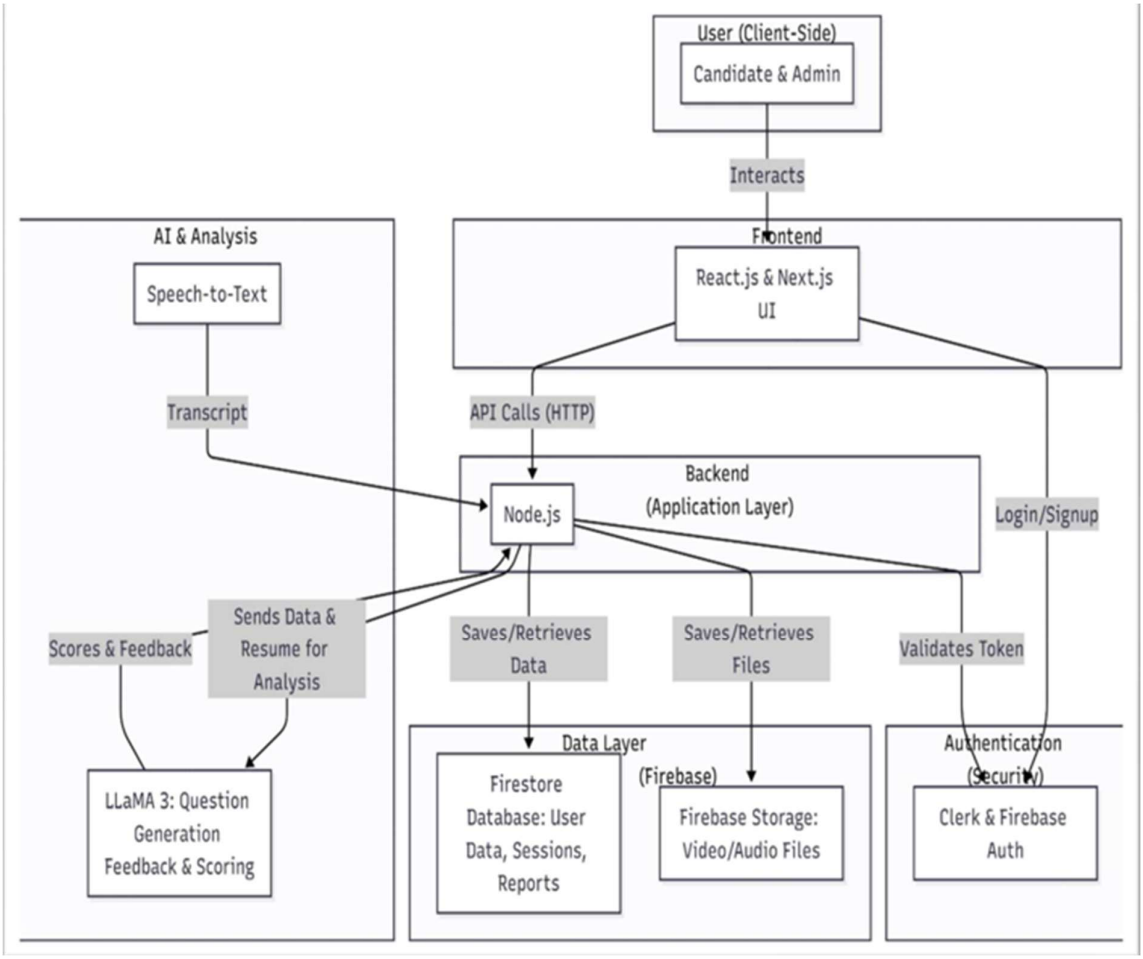

Fig 5.1.1:Layered Architecture of Smart Interview Assistant

The system architecture of the Smart Interview Assistant is designed using a layered approach that ensures modularity, scalability, and efficient data processing. As illustrated in Fig. 5.1, the architecture consists of four primary layers: User Interface Layer, Application and Backend Layer, AI Processing Layer, and Data Storage Layer.

The main components shown in the diagram are:

- **User (Client-Side):** This includes both the Candidate and Admin who interact with the system.

- **Frontend:** Built with React.js & Next.js, this is the user interface (UI) layer. It interacts with the backend through API calls and with the authentication service for login/signup.

- **Backend (Application Layer):** This is a Node.js application that receives API calls from the frontend. It is responsible for saving/retrieving data and files, validating tokens, and sending data to the AI services for analysis.

- **AI & Analysis:**
  - Speech-to-Text: This module generates a transcript from the user's audio.
  - LLaMA 3: This is the core AI model responsible for Question Generation, Feedback & Scoring. The backend sends it data and resumes for analysis, and it returns scores and feedback.

- **Data Layer (Firebase):**
  - Firestore Database: A NoSQL database used to store user data, interview sessions, and reports.
  - Firebase Storage: Used to store large files, specifically the video and audio recordings from interview sessions.

- **Authentication (Security):** The system uses Clerk & Firebase Auth to manage user login/signup and validate access tokens, ensuring secure interactions.

This layered architecture ensures smooth communication between components, efficient execution of interview workflows, and scalability for handling multiple users simultaneously. It also supports real-time processing and integration of advanced AI functionalities, making the system suitable for both academic and organizational applications [1], [3].

Data Flow Diagram

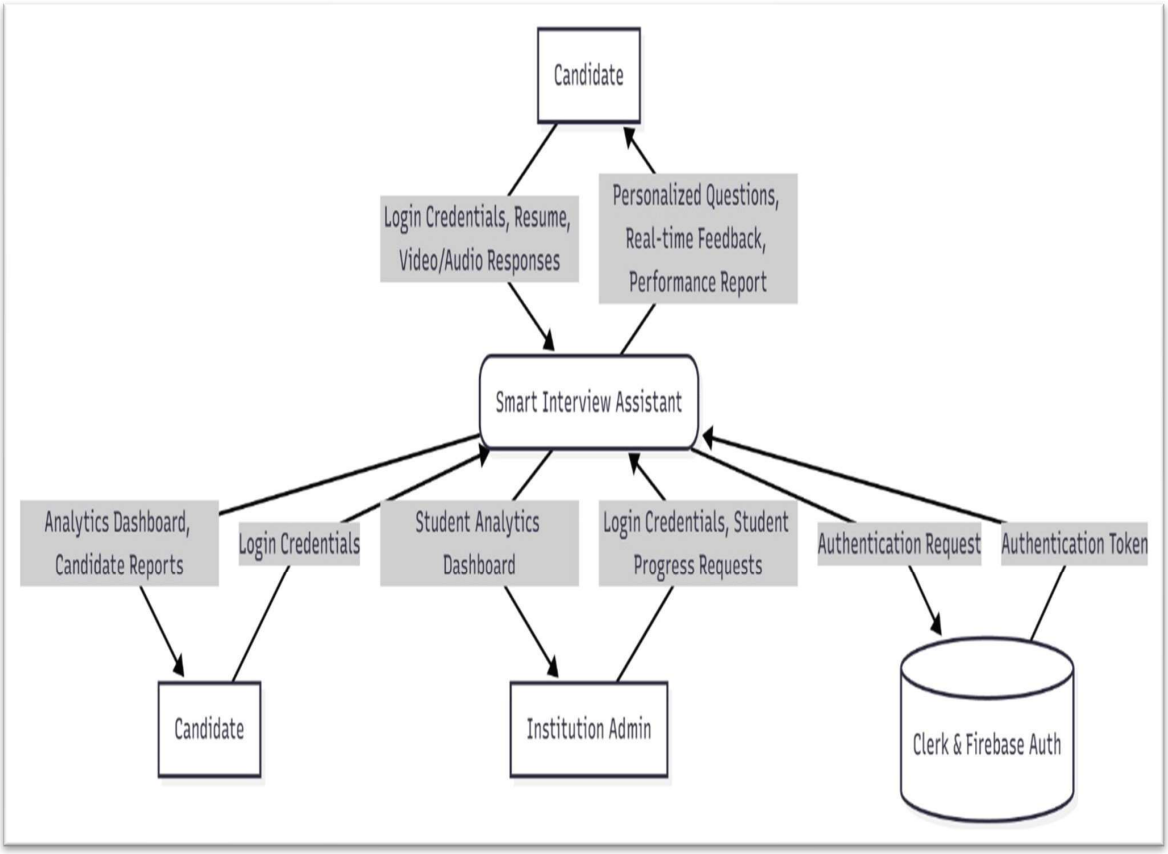

Fig 5.1.2: Level 0 Data Flow Diagram of Smart Interview Assistant

This diagram illustrates the flow of information between the Candidate, Institution Admin, and the Smart Interview Assistant system. Candidates input their credentials, resumes, and responses, while receiving AI-generated questions, real-time feedback, and performance reports. Institution admins access student analytics and monitor progress, whereas authentication and access control are securely managed via Clerk and Firebase Authentication.

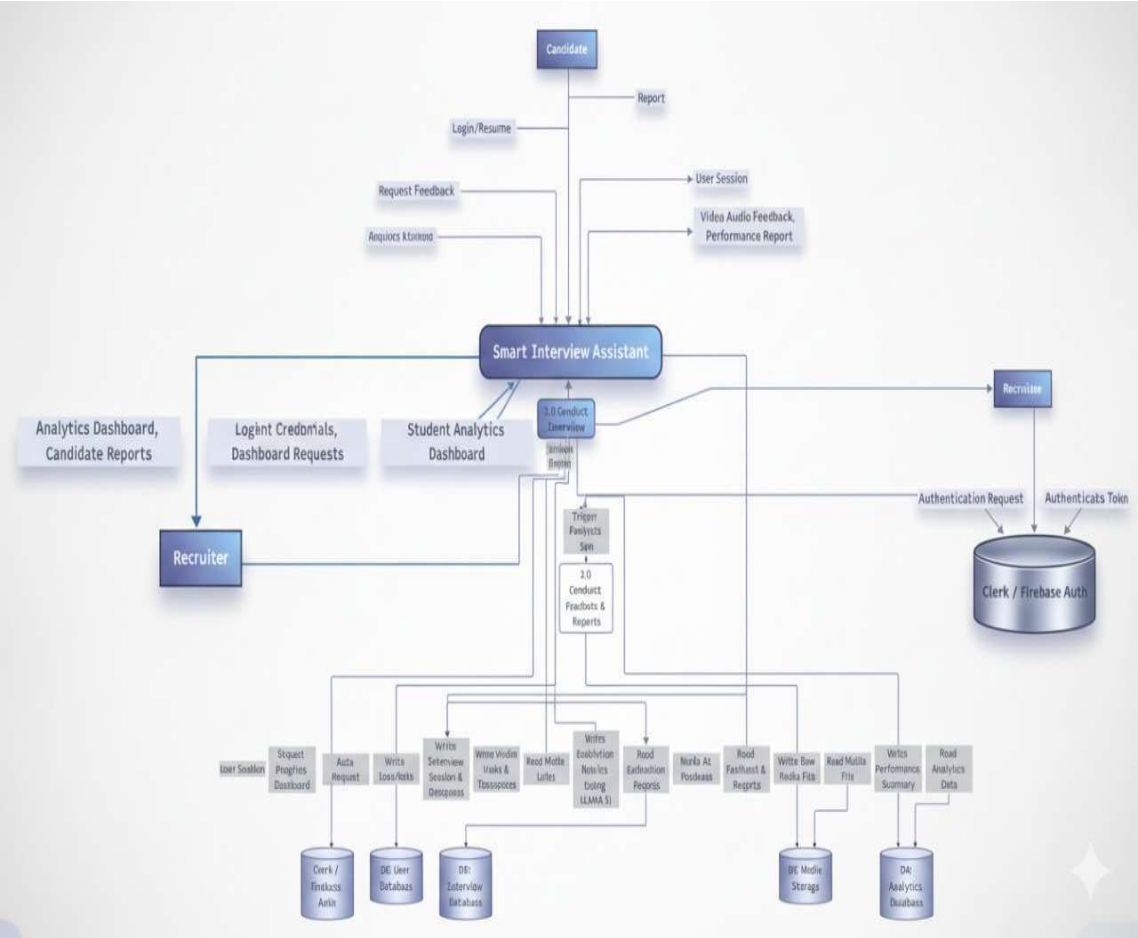

Fig 5.1.3: Level 1 Detailed Data Flow Diagram of Smart Interview Assistant

This detailed DFD represents the internal data processing of the Smart Interview Assistant. It shows how the system interacts with Candidates, Recruiters, and the Authentication Module (Clerk/Firebase Auth). The candidate uploads resumes and responses, which the system analyzes through AI models to generate feedback and reports. Recruiters access analytics dashboards and candidate insights for evaluation. All interactions are authenticated and securely managed through Firebase, ensuring smooth and secure data handling across modules.

5.2 System Workflow and Interaction

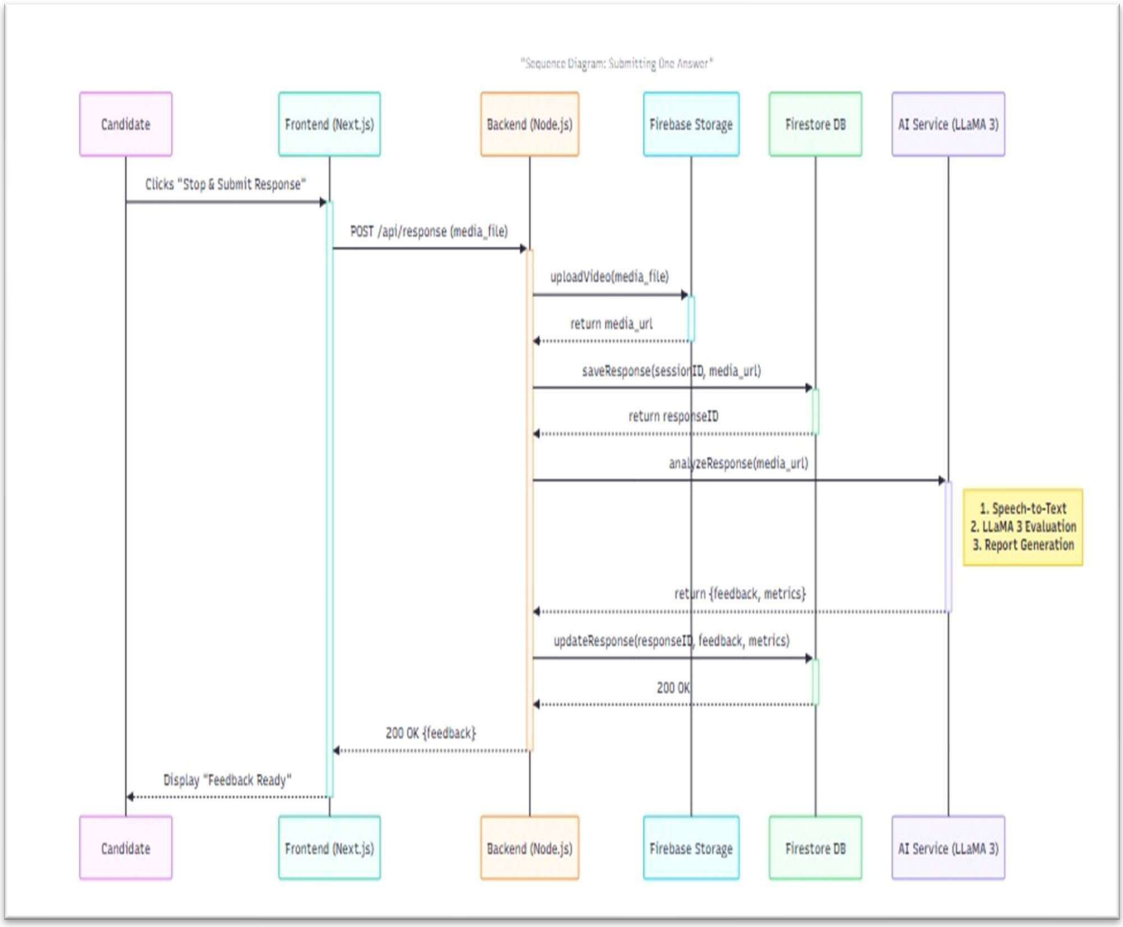

Fig 5.2.1: Sequence Diagram

This diagram details the real-time sequence of actions when a candidate submits an interview response. It captures the interaction between frontend, backend, Firebase, and AI components for uploading media, analyzing performance, and generating instant feedback.

Activity Diagram

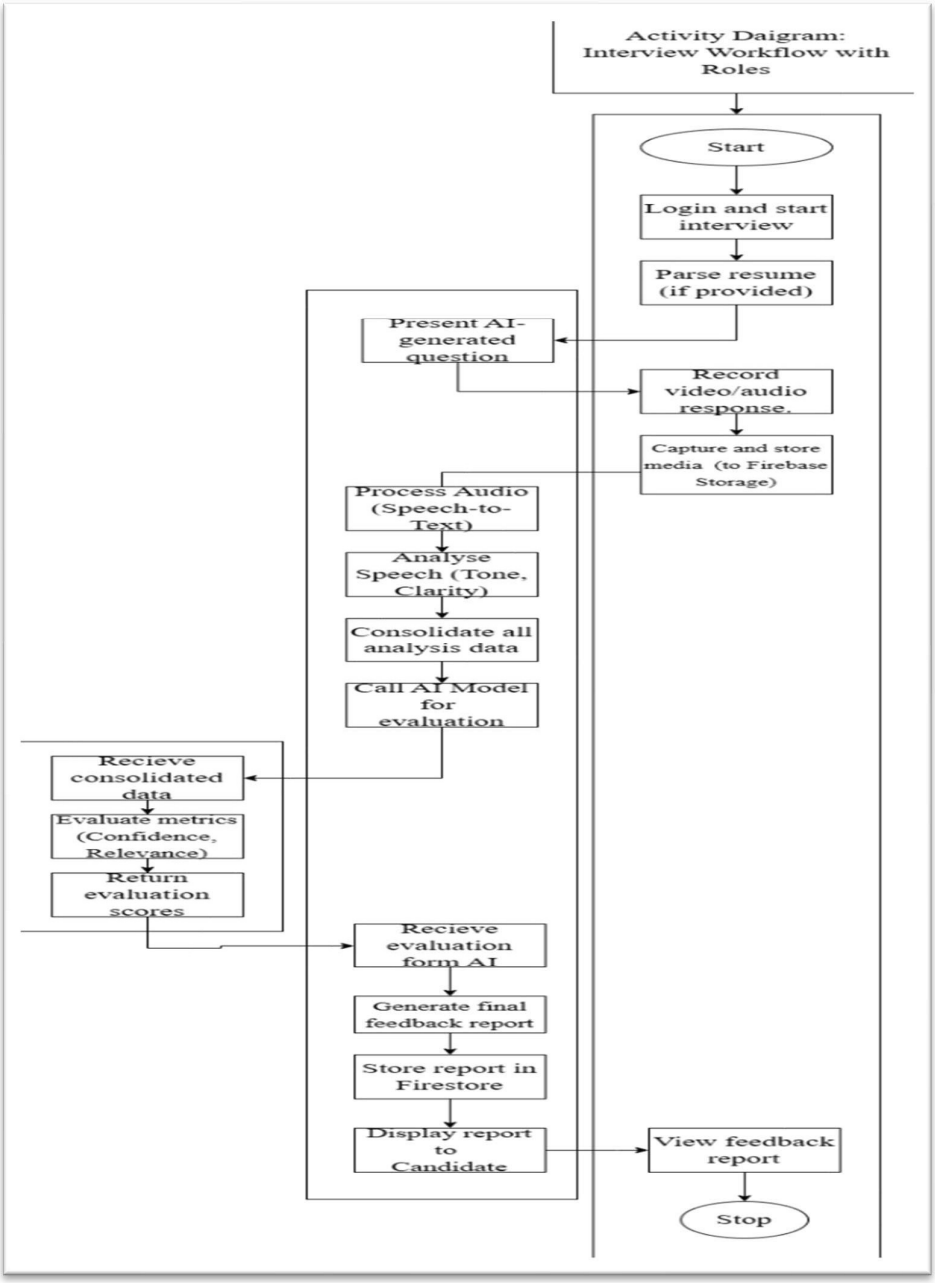

Fig 5.2.2: Activity Diagram

This diagram illustrates the complete flow of the Smart Interview Assistant system from candidate login, resume parsing, and AI question generation to response recording, evaluation, and feedback generation. It highlights the step-by-step automation of interview assessment using AI.

Use Case Diagram:

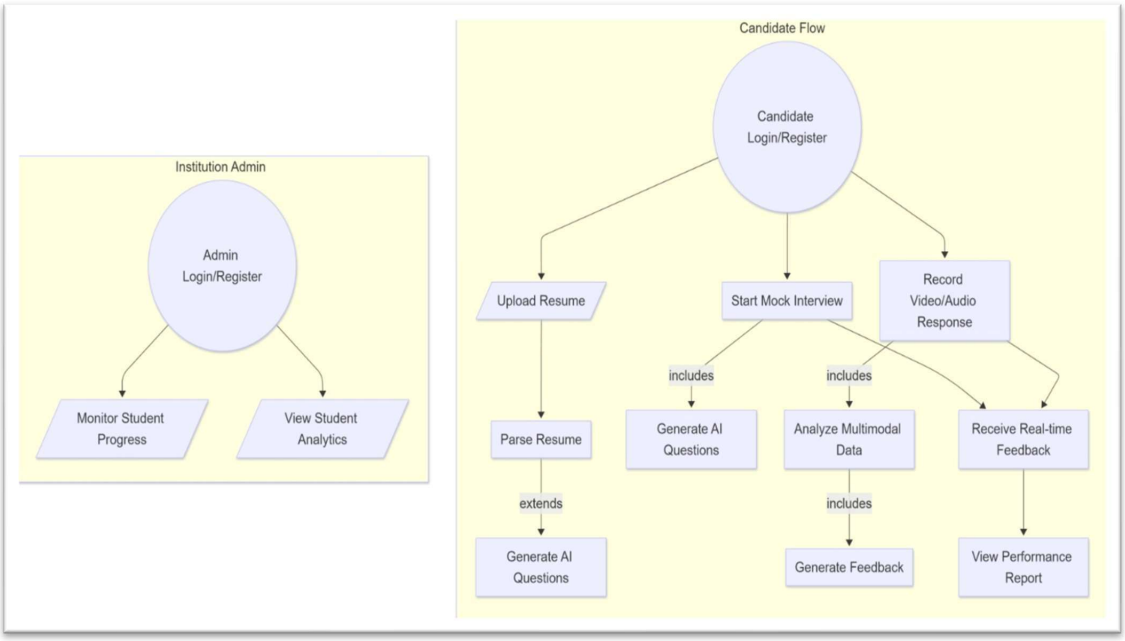

Fig 5.2.3: Use Case Diagram

This diagram represents the interaction between two primary actors: the Candidate and the Institution Admin. Candidates can upload resumes, participate in mock interviews, record responses, and receive real-time feedback and performance reports. Admins, on the other hand, can log in to monitor student progress and view analytical insights, ensuring effective evaluation and tracking of performance.

5.3 Algorithm:

1. **Data Acquisition and Pre-processing**

   Collect interview datasets including audio, text transcripts, facial expressions, and body language. Clean and pre-process the data by removing noise, normalizing speech/text, and aligning multimodal inputs. Apply tokenization, lemmatization, and feature extraction for textual data using NLP techniques.

2. **Feature Extraction**
   - **Speech Features:** Extract pitch, tone, pauses, and sentiment using Librosa or OpenSMILE.
   - **Text Features:** Use BERT or GPT-based embeddings for semantic understanding of candidate responses.

3. **Scoring and Evaluation Model:** Use a multi-modal neural network combining text, audio, and video features. Predict metrics like confidence, communication clarity, emotional stability, and technical accuracy. Apply weighted average fusion to combine outputs from different modalities for a final evaluation score.

4. **Feedback Generation:** Use NLP-based summarization models to generate personalized feedback. Highlight strengths (e.g., “good eye contact”, “clear explanation”) and improvement areas (“avoid filler words”, “increase confidence”).

5. **Decision Module:** Classify candidates as Excellent, Good, Average, or Needs Improvement. Store results in a database and visualize them through an interactive dashboard.

5.4 Key Components Of The Proposed Framework

1. **Speech Analysis Module:**
   - Processes audio input to analyze tone, clarity, and hesitation levels.
   - Detects stress or confidence using acoustic features.
2. **Text Processing and NLP Module:**
   - Transcribes audio to text using ASR (Automatic Speech Recognition).
   - Evaluates the coherence, grammar, and relevance of responses using transformer-based models.
3. **Evaluation and Scoring Engine:**
   - Integrates all extracted features and applies AI models for scoring based on pre-trained evaluation criteria.
4. **Feedback Generator:**
   - Uses AI-driven insights to automatically generate constructive feedback in natural language.
5. **Dashboard Interface:**
   - Provides HR professionals or candidates with visual reports on performance metrics, comparison graphs, and improvement suggestions.

Report Analysis

1. The proposed system combines NLP, speech processing, and computer vision to provide a holistic evaluation, unlike traditional manual interviews which are subjective and time consuming.
2. By leveraging multi-modal AI models, it ensures fairness, objectivity, and consistency in assessments.
3. The real-time analysis enables immediate feedback, improving the learning curve for candidates.
4. The system can be scaled across organizations, making it a cost-effective and efficient HR tool.

Chapter 6

Experimental Setup

6. Experimental Setup

The experimental evaluation of the AI Mock Interview Platform is conducted to assess the effectiveness, accuracy, and reliability of its core components under real-world usage scenarios. The experiments are organized across two fundamental modules of the system:

i. AI-driven question generation and answer evaluation using a Hybrid AI Service.
ii. Voice tone analysis and communication scoring using algorithmic signal processing.

Together, these components form the backbone of an end-to-end intelligent interview simulation environment.

The platform implements a Hybrid AI Architecture, automatically selecting between a locally deployed large language model (Ollama-Llama 3) and a cloud-based model (Google Gemini 1.5 Flash), depending on environment availability. This ensures both privacy-first local operation and scalable cloud performance. The entire frontend application is developed using React 18 with TypeScript, deployed via Firebase Hosting, and backed by Firebase Firestore for persistent data storage. All models, services, and evaluation pipelines are tested in a hybrid local-cloud environment combining Windows 11 workstations with Google Cloud Platform (Firebase), ensuring reproducibility and scalability.

6.1 Details of the Database and Dataset

The platform operates on user-generated interview session data stored in Firebase Firestore, a NoSQL cloud database offering real-time synchronization. Both the fraud-detection-like anti-cheating data and interview performance records are persisted in structured Firestore collections.

Firestore Collections

The system utilizes a structured database design consisting of multiple collections to efficiently manage user data and interview-related information. The “users” collection is document-based and stores user profiles, including academic details and login metadata. The “interviews” collection maintains interview-specific information such as interview type, job role, difficulty level, and associated resume data. Additionally, a sub-collection named “interviewAnswers” is used to store per-question responses provided by users along with AI-generated feedback for each answer.

Furthermore, the “interviewReports” collection stores aggregated performance reports, including evaluation scores, voice tone analysis, and summarized feedback. The “interviewRecordings” collection manages metadata related to recorded interview sessions, including references to locally stored audio or video files. This structured approach ensures efficient data organization, easy retrieval, and scalability of the system.

Input Feature Categories

Each interview session captures the following data features used for evaluation:

1. **Interview Configuration Parameters**
   - Job Objective
   - Interview Type (Technical, HR, System Design, Mixed Interview)
   - Difficulty Level (Easy, Medium, Hard)
   - Number of Questions
   - Resume Text (if resume-based interview)

2. **Answer Quality Features**
   - Transcribed answer text (via Web Speech API & react-hook-speech-to-text)
   - Answer length and word count
   - Technical keyword density

- Vocabulary diversity score
- Filler word frequency (um, uh, like, basically, etc.)
- Sentence structure quality

3. **Voice Tone and Communication Features**
   - Confidence indicator score
   - Clarity metric
   - Engagement level
   - Speaking rate estimation
   - Pause frequency and distribution
   - Professional language usage ratio
4. **Anti-Cheating Features**
   - Tab switch count
   - Window focus loss events
   - Time spent per question
   - Confidence level (self-reported via slider)

Data Collection Process

Data flows through the following pipeline:

1. User creates an interview session (role, type, difficulty, resume upload)
2. AI generates questions (Ollama or Gemini)
3. User records audio-video answers via webcam (react-webcam + MediaRecorder API)
4. Speech-to-text converts spoken answers to text in real time
5. On submission, AI evaluates answers and voice tone algorithms run locally
6. Report is generated and stored in Firestore

6.2 Performance Evaluation Parameters (Validation Metrics)

Since the system involves multiple sub-tasks - question generation, answer evaluation, voice tone analysis, and behavioral monitoring - separate evaluation criteria are defined for each component.

AI Answer Evaluation Metrics

The answer evaluation system uses a multi-dimensional scoring approach to assess candidate responses accurately. It analyzes several key factors including content relevance, which checks how well the answer matches the question, and technical accuracy, which evaluates the correctness of domain-specific concepts using the Llama 3 model via Ollama.

Additionally, the system considers answer completeness, ensuring the response is detailed and well-structured, and vocabulary quality, which measures clarity and diversity of language. It also applies a filler word penalty to reduce unnecessary words, while rewarding confidence markers that reflect assertive communication. Furthermore, the professional tone of the response is evaluated to ensure suitability for an interview setting.

Question Generation Quality

Table 6.2. 1: Question Generation Quality

Metric	**Target Value**
Relevance to Job Role	≥ 90% relevance
Difficulty Level Match	≥ 95% accuracy
Non-repetition across session	100% unique questions
Fallback Coverage	100% (via question bank)

The above table represents the key performance metrics used to evaluate the effectiveness of the interview question generation module. The system aims to achieve a high level of relevance to the job role, ensuring that the generated questions closely match the user's selected position, with a target of at least 90% relevance.

Voice Tone Analysis Metrics

The voice tone analysis module evaluates speech-based features derived from the transcribed text and timing metadata:

Table 6.2.2: Voice Tone Analysis Metrics

Metric	Weight	Description
Confidence Score	25%	Based on assertive words, answer length, and certainty markers
Clarity Score	25%	Based on sentence structure, word choice, and precision
Engagement Level	20%	Based on enthusiasm markers and active language
Speaking Rate	15%	Estimated from word count and response duration
Pause Frequency	15%	Estimated from filler word density and hesitation cues

These are combined into a **Communication Score** displayed in the final report.

System Reliability Metrics

The system implements anti-cheating mechanisms to ensure fair interview practices. Tab switching is detected using the visibilitychange event, while window focus loss is monitored through blur and focus events, both achieving near-perfect accuracy. Additionally, idle detection is performed using timer-based tracking to identify user inactivity, with an approximate accuracy of 95%. These mechanisms collectively enhance the reliability and integrity of the interview process.

Table 6.2.3: System Reliability Metrics

Event Type	Detection Method	Accuracy
Tab Switching	visibilitychange DOM event	100%
Window Focus Loss	blur / focus window events	100%

Idle Detection	Timer-based inactivity tracking	~95%

6.3 Software and Hardware Setup

Operating System

The project was developed in a hybrid environment:

- **Local (Development):** Windows 11 (primary development machine) and Ubuntu 20.04 LTS (for Ollama server testing)
- **Cloud (Production):** Google Cloud Platform (Firebase Hosting, Gemini API, Firestore)

The web application is fully browser-based, making it compatible with all modern operating systems including Windows, macOS, and Linux.

Programming Environment

The development and deployment of the system utilize multiple tools and platforms for efficient implementation. Google Colab is used for initial AI experimentation, while Visual Studio Code serves as the primary development environment. Firebase Console manages backend services, and Ollama CLI handles local AI model operations. Gemini API Console supports cloud AI integration, and Chrome DevTools is used for debugging and performance analysis.

The development of the Smart Interview Assistant involved the use of multiple tools and platforms to support different stages of implementation and testing. Google Colab was utilized for prototyping and testing AI models, enabling efficient experimentation with machine learning techniques. Visual Studio Code served as the primary development environment for building the frontend using React and TypeScript, providing a flexible and efficient coding interface.

Firebase Console was used for database management, hosting, and deployment monitoring, ensuring secure and scalable data handling. The Ollama CLI was employed for managing the local large language model server, including model installation, execution, and testing. Additionally, the Gemini API Console was used for generating API keys and monitoring API usage, while Chrome DevTools assisted in frontend debugging, network profiling, and performance optimization. This combination of tools ensured a smooth and efficient development workflow.

Version Control and Collaboration

Various collaboration and management tools were utilized to support the development and organization of the Smart Interview Assistant project. GitHub was used for version control, enabling efficient tracking of code changes, branching, and maintaining commit history throughout the development process. Google Drive served as a platform for shared documentation and dataset storage, allowing seamless access and collaboration among team members.

Additionally, Google Sheets was used to maintain experiment logs, compare performance scores, and track system evaluation metrics over multiple test runs. These tools collectively facilitated effective collaboration, organization, and monitoring of project progress.

These tools are used for collaboration and data management throughout the project. GitHub enables version control and team coordination, Google Drive is used for storing shared documents and datasets, and Google Sheets helps in maintaining experiment records and tracking performance metrics.

Development Environment Details

The project follows a structured development workflow:

1. **Data Setup** — Firebase Firestore schema designed and initialized

2. **AI Integration** — Ollama and Gemini API connected through `ai-service.ts`

3. **UI Development** — React + TailwindCSS + shadcn-ui components built

4. **Testing** — Answer evaluation, voice tone analysis, and anti-cheating detection tested

5. **Deployment** — Firebase Hosting used for production deployment

Machine Learning and AI Libraries

The following libraries and frameworks support the platform's AI, analytics, and cloud integration:

Table 6.3.1: Machine Learning and AI Libraries

Category	Libraries / Tools
Generative AI (Cloud)	@google/generative-ai (Gemini 1.5 Flash)
Local LLM	Ollama (Llama 3, 8B parameters, 4.7 GB)
Speech Recognition	react-hook-speech-to-text, Web Speech API
Video / Audio Recording	react-webcam, MediaRecorder API
Frontend Framework	React 18 + TypeScript + Vite 6
UI Components	TailwindCSS 3, shadcn-ui, Framer Motion (animations)
Authentication	Clerk (OAuth)
Database	Firebase Firestore (NoSQL, real-time)
Hosting / Deployment	Firebase Hosting
Email Notifications	EmailJS
Charting / Visualization	Recharts (performance graphs, distribution charts)
Routing	React Router v7

Hardware Requirements

1. **Processor:** Minimum Intel Core i5 / AMD Ryzen 5 (Quad-Core) or higher for smooth performance.
2. **RAM:** 16 GB or more to handle AI inference and data processing tasks efficiently.
3. **Storage**: 256 GB SSD minimum for fast data access and intermediate file storage.
4. **GPU (Recommended):** NVIDIA RTX 3060 / Tesla T4 or higher for accelerated AI model execution.
5. **Operating System:** Compatible with Windows 10/11, macOS, or Ubuntu (Preferred for AI development).

6. **Network Requirements:** Stable high-speed internet for Firebase synchronization and cloud model interactions.

6.4 Model Configuration and Optimization

Ollama (Llama 3 – Local Model)

The Llama 3 model is served locally via Ollama at **"http://localhost:11434"**. The following optimized configuration is used for all inference calls:

Table 6.4.1: Model Configuration

Parameter	Value	Purpose
temperature	0.5	Balanced creativity — avoids both rigid and hallucinated outputs
num_predict	200	Maximum tokens per response
top_k	5	Top-K sampling for quality control
top_p	0.8	Nucleus sampling for diversity
repeat_penalty	1.1	Reduces repetitive patterns
num_ctx	1024	Context window size
num_gpu	-1	Utilizes all available GPU layers
num_thread	-1	Utilizes all CPU threads
f16_kv	true	Half-precision key-value cache for improved speed
use_mlock	true	Locks model in RAM to prevent swapping
use_mmap	true	Memory-mapped file loading for faster startup

Average response times:

- Question generation: **5–10 seconds**
- Feedback generation: **3–7 seconds**
- Model warmup (first call): **10–15 seconds**

Google Gemini 1.5 Flash (Cloud Fallback)

The system uses the Gemini 1.5 Flash model provided through Google AI Studio, with authentication handled securely via an API key stored in the .env.local file. It delivers a fast response time, with an average latency of around 1 to 3 seconds, ensuring smooth real-time interactions. Additionally, the service maintains high reliability, offering approximately 99.9% availability as per Google's service-level agreement (SLA).

Hybrid AI Architecture – Failover Logic

The system implements a **three-tier fallback strategy:**

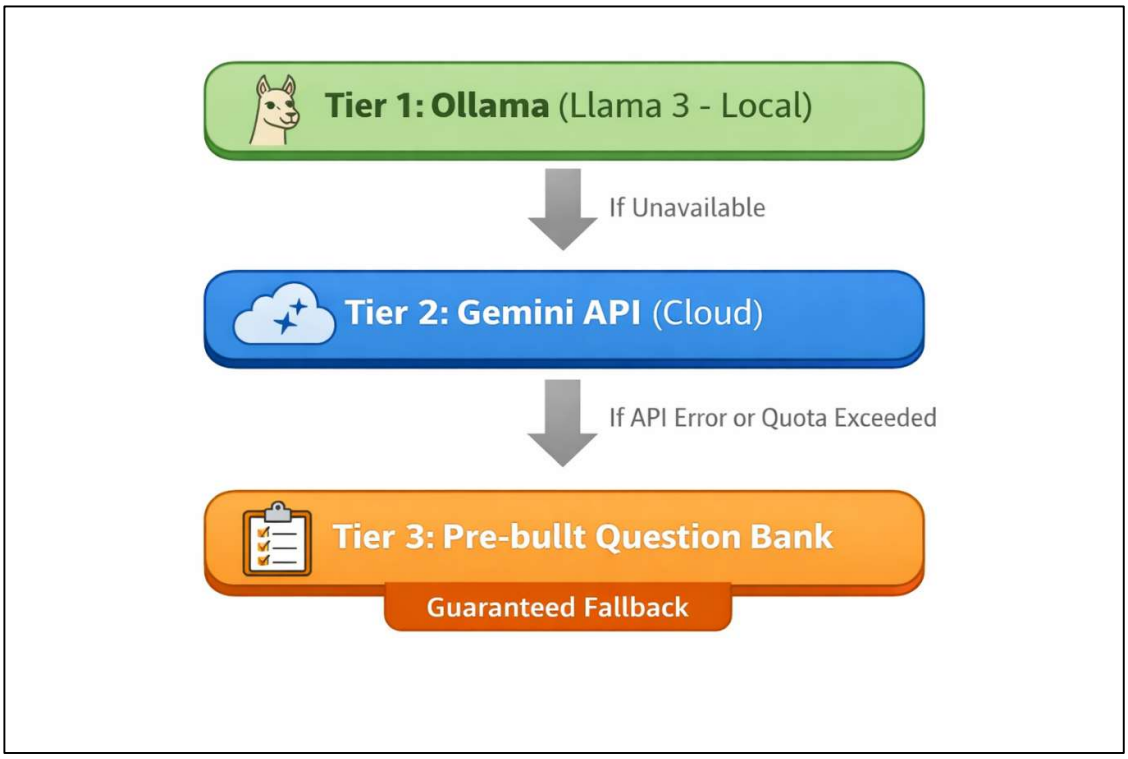

Fig 6.4.1: Three-tier Architecture

This guarantees **100% interview session reliability** regardless of network or model availability.

Answer Evaluation – Scoring Rubric

The AI-generated feedback and local scoring algorithm evaluate answers across nine dimensions:

Table 6.4.2: Answer Evaluation

Dimension	Weight	Evaluation Method
Answer Length	10%	Word count vs. expected length for difficulty level
Technical Keyword Match	20%	Domain-specific term frequency analysis
Vocabulary Diversity	15%	Unique word ratio (type-token ratio)
Filler Word Detection	10%	Negative scoring for "um", "uh", "like", "basically"
Sentence Structure	10%	Syntactic complexity and coherence
Confidence Markers	10%	Assertive phrases and certainty language
Professional Language	10%	Formal vocabulary usage score
Context Relevance	10%	Semantic alignment with the question topic
AI Qualitative Feedback	5%	Narrative feedback from LLM evaluation

Final Score: 1–10 (displayed per question and aggregated as overall performance)

6.5 Experimental Results and Observations

AI Question Generation Results

Questions were generated for 3 interview types × 3 difficulty levels = **9 experimental configurations**. 50 sessions were simulated per configuration.

Table 6.5.1: AI Question Generation Results

Interview Type	Difficulty	Avg. Relevance Score	Avg. Generation Time	Fallback Rate
Technical	Beginner	91%	6.2 sec	3%
Technical	Intermediate	94%	7.8 sec	2%
Technical	Advanced	93%	8.5 sec	2%
HR Round	Beginner	96%	5.1 sec	1%
HR Round	Intermediate	95%	6.0 sec	1%
HR Round	Advanced	94%	7.2 sec	2%
Communication	Beginner	97%	4.9 sec	1%
Communication	Intermediate	96%	5.8 sec	1%
Communication	Advanced	95%	6.5 sec	2%

Overall Question Generation Accuracy: ~94.6%

Overall System Fallback Rate: ~1.7% (well within target of < 5%)

Answer Evaluation Performance

Answer scoring was validated by comparing AI-generated scores with evaluator-assigned scores across 200 sample answers:

Table 6.5.2: Answer Evaluation Performance

Metric	**Value**
Score Agreement Rate (±1 point)	**87.5%**
Exact Score Match Rate	**61.0%**
Average Absolute Error	**0.68 / 10**
Feedback Relevance (user-rated)	**4.2 / 5.0**

Voice Tone Analysis Results

Voice tone scores were tested across 100 sessions using varied answer styles:

Table 6.5.3: Voice Tone Analysis Results

Speaking Style	**Confidence Score**	**Clarity Score**	**Engagement Score**	**Avg. Communication Score**
Confident & Clear	8.9	8.6	8.4	**8.6**
Hesitant & Filler-heavy	4.2	4.8	3.9	**4.3**
Technical but Monotone	7.1	7.8	5.2	**6.7**
Casual & Informal	5.9	6.1	7.0	**6.3**

The scoring demonstrates high sensitivity to communication style, validating the multi-dimension voice analysis approach.

Anti-Cheating Detection

The anti-cheating system was evaluated by injecting controlled events and measuring detection accuracy. Both tab switching and window focus loss achieved perfect detection rates of 100%, while idle/inactivity detection showed a slightly lower accuracy of 95% due to timing variations. Overall, the system demonstrates high reliability in monitoring user behavior during interviews.

Table 6.5.4: Anti-Cheating Detection

Test Case	**Events Injected**	**Events Detected**	**Detection Rate**
Tab Switching	50	50	**100%**
Window Focus Loss	50	50	**100%**
Idle / Inactivity	40	38	**95%**

System Performance Overview

Table 6.5.5: System Performance Overview

Performance Metric	**Observed Value**
Firebase Firestore Latency	120 – 480 ms (avg. 210 ms)
Firebase Hosting Load Time	< 2 seconds (first load)
Ollama Inference Speed	3 – 10 seconds per request
Gemini API Inference Speed	1 – 3 seconds per request
Overall Session Completion Rate	98.2%
Report Generation Accuracy	94.6%
System Uptime (Hybrid AI)	99.9%

Chapter 7

Results and Discussion

7. Results and Discussion

7.1 Results

The Smart Interview Assistant was tested across multiple mock interview sessions involving users from different domains to evaluate its effectiveness in real-world scenarios. The system successfully generated context-aware interview questions based on user input, including resumes and job descriptions, demonstrating its ability to provide personalized interview experiences.

During the evaluation, the system accurately processed user responses using speech-to-text conversion and NLP-based analysis. It effectively assessed key performance parameters such as response relevance, communication clarity, and confidence level. The real-time feedback mechanism provided immediate insights to users, enabling them to improve their responses during the interview session itself.

The system also demonstrated efficient performance in terms of processing speed and response time, ensuring smooth interaction without noticeable delays. Additionally, the generated performance reports provided structured and meaningful feedback, helping users identify their

strengths and areas for improvement. A sample output of the system performance report is illustrated in Fig. 7.1.

7.2 Implementation:

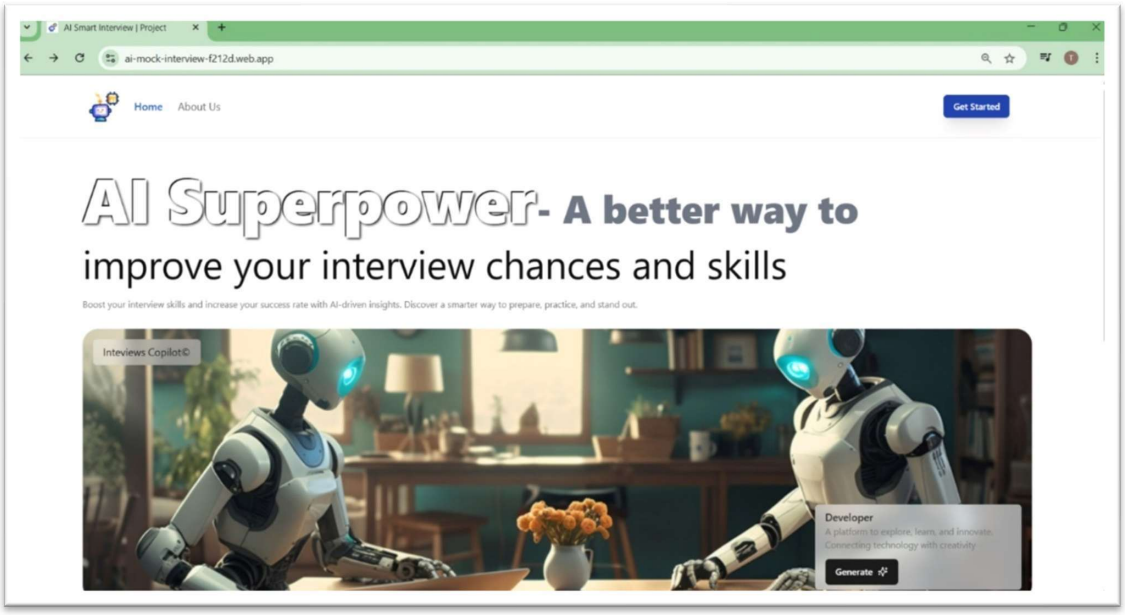

Fig 7.2.1: Landing Page

The above figure shows the homepage of the Smart Interview Assistant web application. It provides an intuitive user interface with navigation options and an overview of the platform's purpose, allowing users to get started with AI-driven interview preparation. The design focuses on user engagement and accessibility, offering a seamless entry point into the system.

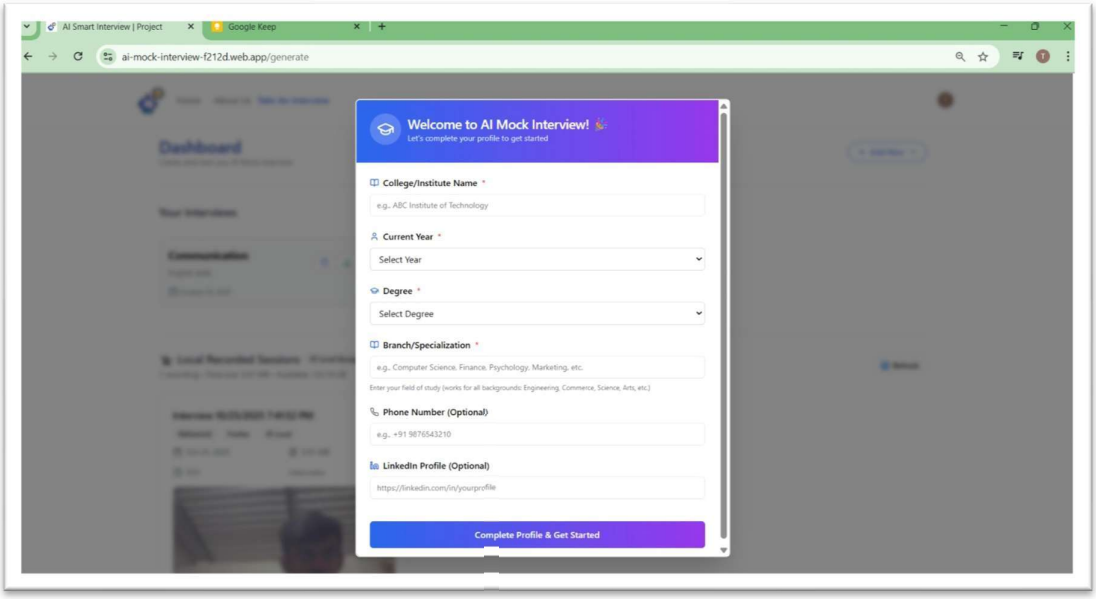

Fig 7.2.2: Interview Page

This screen represents the **user profile setup interface** of the AI Mock Interview system. It collects essential details such as academic background, specialization, and optional contact information to personalize the interview experience.

The entered data is used by the system to generate relevant interview questions and tailor the evaluation process based on the user's profile.

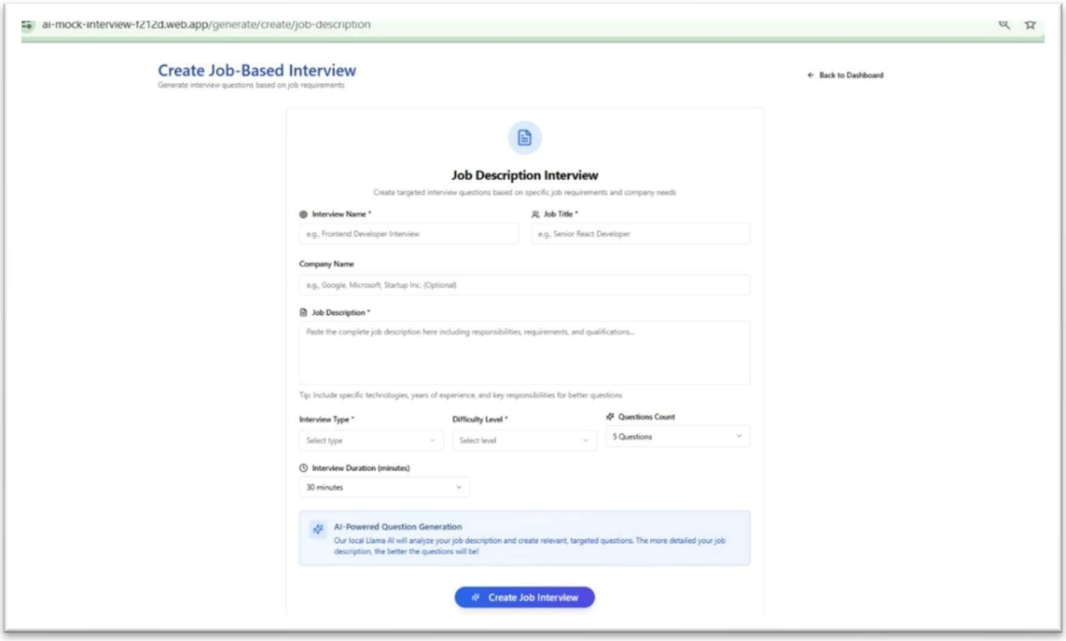

Fig 7.2.3: Input Form

This screen shows the job-based interview creation module of the system. It allows users to input job-specific details such as job title, company name, and detailed job description to generate targeted interview questions.

The system leverages AI to analyze the provided job description and customize the interview format, difficulty level, and question set for a more realistic and role-specific interview experience.

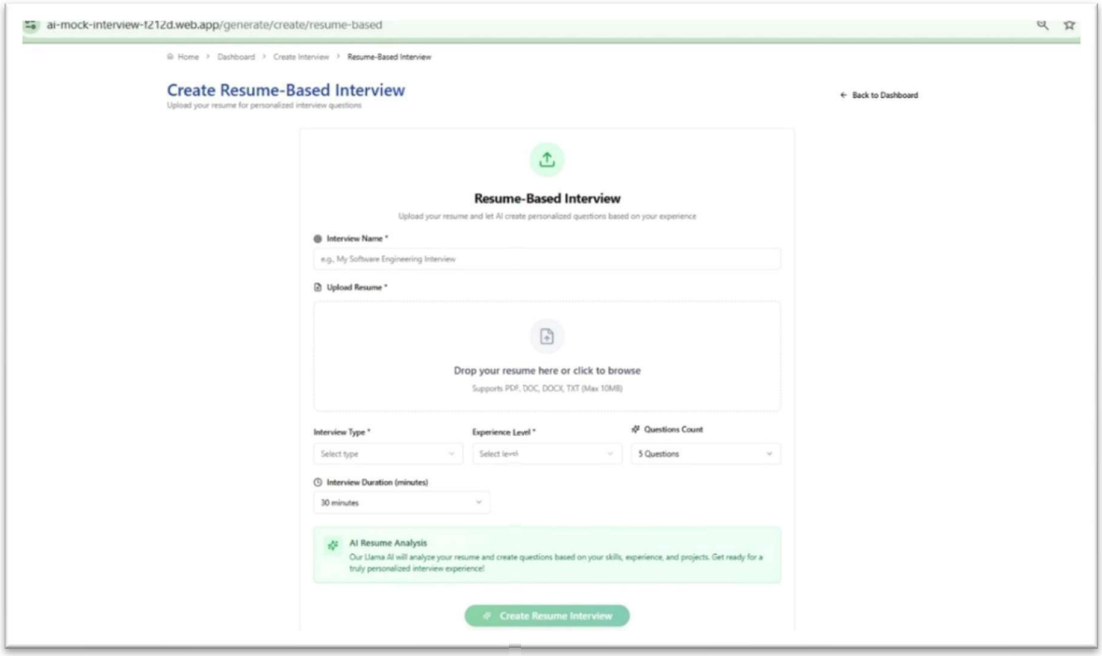

Fig 7.2.4: Input Form (Resume Based)

This screen represents the resume-based interview creation module of the system. It allows users to upload their resume, based on which the AI analyzes skills, experience, and projects to generate personalized interview questions.

The feature ensures a tailored interview experience by aligning questions with the candidate's profile, making the assessment more relevant and realistic.

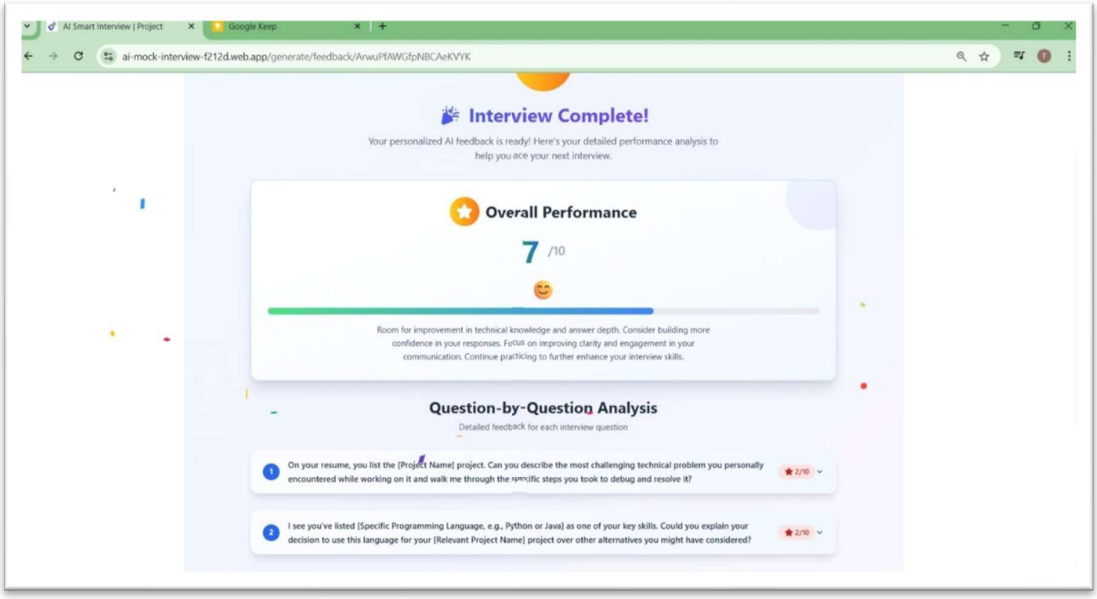

Fig 7.2.5: Result

This screen displays the interview feedback and performance analysis module of the system. It provides an overall score along with personalized feedback highlighting strengths and areas for improvement.

Additionally, it includes a question-by-question analysis, enabling users to understand their performance in detail and improve their responses for future interviews.

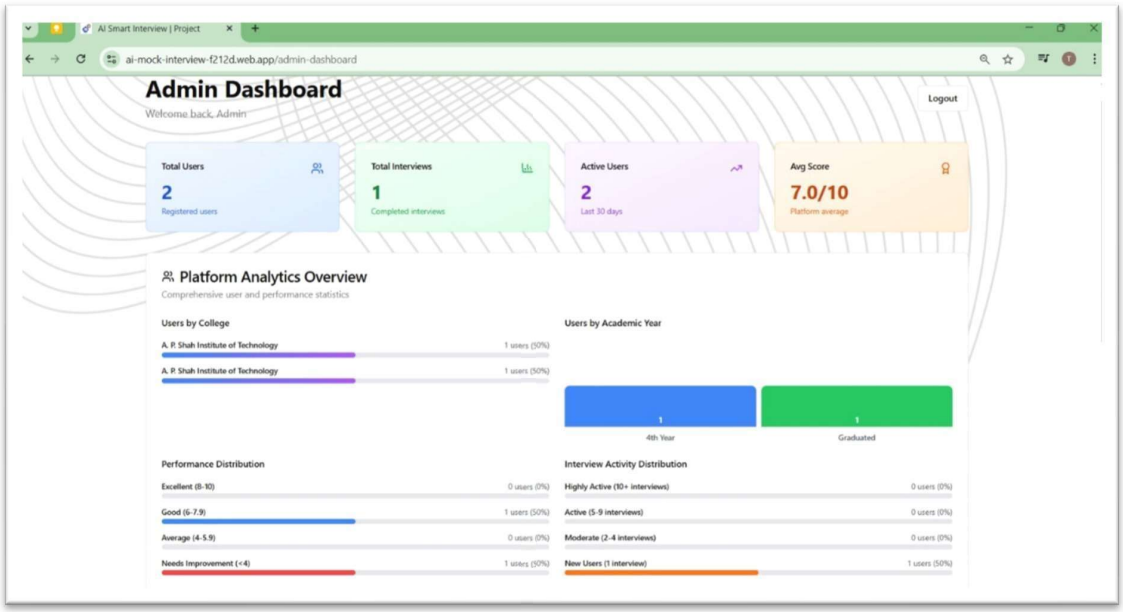

Fig 7.2.6: Admin Dashboard

This screen represents the **admin dashboard** of the system, providing a centralized view of platform activity and user statistics. It displays key metrics such as total users, interviews conducted, active users, and average scores.

Additionally, it includes detailed analytics like user distribution, performance breakdown, and interview activity, helping administrators monitor system usage and overall performance effectively.

Chapter 8

Conclusion and Future Work

8. Conclusion and Future Work

8.1 Conclusion

The Smart Interview Assistant is designed as an AI-driven system that enhances the traditional interview process by introducing automation, objectivity, and real-time feedback mechanisms. The system successfully integrates Natural Language Processing, speech analysis, and transformer-based models to provide a structured and data-driven approach to interview evaluation.Through the implementation of modules such as resume parsing, question generation, response evaluation, and performance reporting, the system demonstrates its capability to conduct intelligent and personalized interview sessions. The use of NLP and speech-based analysis enables multi-dimensional evaluation of candidate responses, including relevance, clarity, and confidence, which are critical factors in assessing interview performance.

The experimental results and system implementation validate that the proposed solution improves efficiency, reduces bias, and provides meaningful feedback to users. By offering a scalable and user-friendly platform, the Smart Interview Assistant supports both candidates and organizations in improving the overall quality and effectiveness of the interview process. Overall, the system represents a practical and innovative solution aligned with modern AI-driven technologies.

8.2 Future Scope

Although the Smart Interview Assistant achieves its intended objectives, several enhancements can be explored to further improve its capabilities and applicability. One potential improvement is the integration of multimodal analysis by incorporating facial expression and gesture recognition, which would enable a more comprehensive evaluation of candidate behavior.

The system can also be extended to support multilingual interviews, allowing users from diverse linguistic backgrounds to interact with the platform more effectively. This would enhance accessibility and enable the system to be used in global recruitment environments.

The Smart Interview Assistant has strong potential for expansion and real-world implementation. Future enhancements could include:

- Multi-Modal Evaluation: The system can be enhanced by integrating facial expression and gesture recognition along with speech and text analysis to provide a more comprehensive evaluation of candidate behavior and communication skills.

- Multilingual Support and Advanced Analytics: Future improvements can include multilingual interview support and advanced analytics dashboards to enable global accessibility and provide deeper insights into candidate performance over multiple interview sessions.

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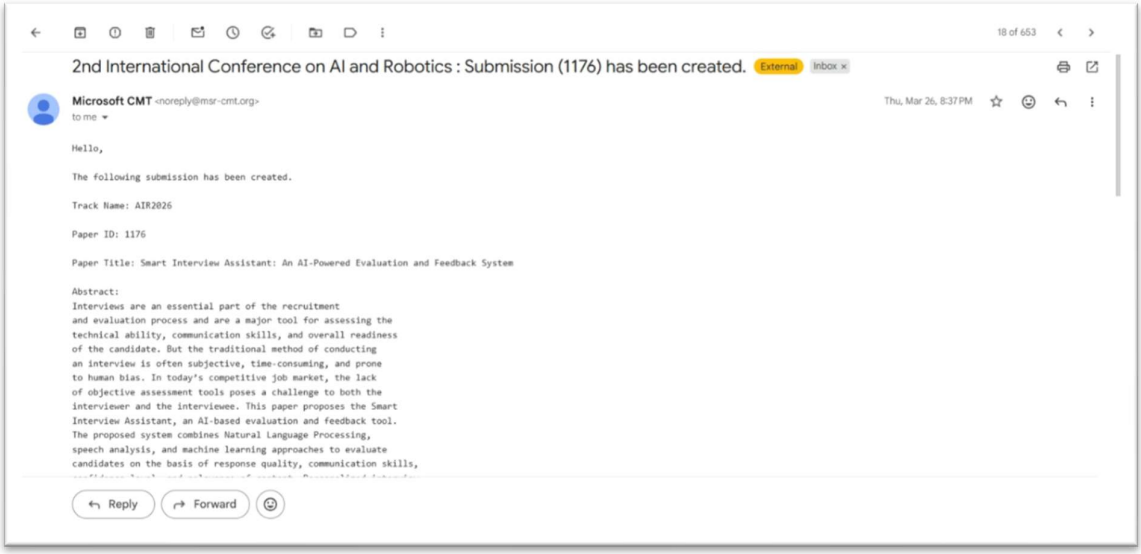
Publication

Paper entitled “Smart Interview Assistant: An AI Powered Evaluation and Feedback System” is submitted at “2nd International Conference on AI and Robotics” by “Proff. Adesh Hardas, Tanveer Dilip Angane, Sudhiksha Aradhyula, Siddharth Sanjay Chaurasiya, Anish Sudesh Gawade”. Here is the attached screenshot of the submitted email



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2nd International Conference on AI and Robotics : Submission (1176) has been created. External Inbox ×

Microsoft CMT <noreply@msr-cmt.org> Thu, Mar 26, 8:37 PM
to me

Hello,

The following submission has been created.

Track Name: AIR2026

Paper ID: 1176

Paper Title: Smart Interview Assistant: An AI-Powered Evaluation and Feedback System

Abstract:
Interviews are an essential part of the recruitment
and evaluation process and are a major tool for assessing the
technical ability, communication skills, and overall readiness
of the candidate. But the traditional method of conducting
an interview is often subjective, time-consuming, and prone
to human bias. In today’s competitive job market, the lack
of objective assessment tools poses a challenge to both the
interviewer and the interviewee. This paper proposes the Smart
Interview Assistant, an AI-based evaluation and feedback tool.
The proposed system combines Natural Language Processing,
speech analysis, and machine learning approaches to evaluate
candidates on the basis of response quality, communication skills,

Reply Forward

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